



Latent trait models for perceived risk assessment using a Covid-19 data survey

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ABSTRACT

Aim of the contribution is analyzing potential events that may negatively impact individuals, assets, and/or the environment, and making judgments about the perceived personal and social riskiness of Covid-19 compared to other hazards belonging to health (AIDS, cancer, infarction), environmental (climate change), behavioral (serious car accidents), and technological (nuclear weapons) domains. The comparative risk analysis has been performed on a survey data collected during the first Italian Covid-19 lockdown. An item response theory model for polytomously scored items has been implemented for the analysis of the positioning of Covid-19 with respect to the other hazards in terms of perceived risk. Among the attributes determining the hazard's perceived risk, Covid-19 distinguishes for the knowledge of risks from the hazard, media attention, and fear caused by the hazard in the peers. Besides, through a latent regression analysis, the role of some individual characteristics on the perceived risk for Covid-19 has been examined. Our contribution allows us to disentangle among several aspects of hazards and describe the main factors affecting the perceived risk. It also contributes to determine if existing control measures are perceived as adequate and the interest for new media with related impact on a person's reaction.

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1. Introduction

During the past years, researchers have been analyzing risk intensively and from many perspectives. The main focus has been on risk assessment and risk management. The former involves the identification, quantification, and characterization of threats to human health and the environment. The latter deals with processes of communication, mitigation, and decision making. Public perception of risk plays a central role in risk analysis, bringing issues of values, process, power, and trust [61]. There are multiple conceptions of risk interpreted as a hazard, probability, and consequence. For several authors, the risk

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is seen as inherently subjective [39,66,75] or denoted as feelings in which affective reactions play the determinant role [65]. In this area, Weber [75] reviewed three approaches by which risk perception has been studied: the axiomatic measurement paradigm, the socio-cultural paradigm, and the psychometric paradigm. Within the psychometric paradigm, people make quantitative judgments about diverse hazards' riskiness, revealing their perspectives. Understanding people's judgment is needed to develop effective public policies since people respond only to the hazards they perceive [64]. Indeed, whereas experts base their judgment about riskiness on technical measurements, laypeople consider other hazard's characteristics in their assessment of risk, e.g. control over the potential damages, harmfulness for children, and irreversibility of effects [65].

About that, psychometric studies have demonstrated that the characteristics of hazards with their perceived risk play a key role in the process known as social amplification of risk [38]. Moreover, the comparison of different hazards by means of their characteristics represents a useful tool for exploring the risk perception construct. Indeed, the human association-based information processing, which works by way of similarity and associations, has a crucial role in people risk perception, transforming uncertain and threatening aspects in the environment into affective responses (e.g. fear, dread, and anxiety) [44]. That is, the hazards people worry about most have some characteristics in common.

The presented analysis moves in this context analyzing risk perception of Covid-19 compared to several other hazards. The Covid-19 global pandemic began in December 2019, and it is still in 2021 a worldwide emergency and probably it will continue to be so in the next few years damaging different areas of people's lives, such as health, work, economy, and relationship. The required containment measures also contribute to people's distress. Several studies investigated the physical and psychological consequences of the Covid-19 emergency, some of them focusing on specific sub-populations as adolescents, health care workers, or students, for instance [20,24,30]. Among others, Commodari and La Rosa [20] discussed the effects of quarantine on Italian adolescents' lifestyle and well-being, reporting that the lockdown had increased their sadness, irritation, and difficulty in concentrating and sleeping. Results also showed that the quarantine had affected adolescents' self-confidence and sense of security. Other examples concerned analyses on depression and anxiety for categories particularly exposed to Covid-19 contagion as the health care workers [30,47]. Son *et al.* [69], instead, underlined the increasing concern related to students' mental health showing that the Covid-19 pandemic situation delivered this vulnerable population into renewed focus.

In this vein, the study of Covid-19 perceived risk is of great interest since it has an important role in pushing people to take appropriate protective behavior [12,35], such as the use of facial masks [34] or the compliance with social distancing measures [29].

In particular, we investigated people's judgment about the perceived risk for the individual (say, personal riskiness) and for the community (say, social riskiness) of Covid-19 compared to other several hazards mainly belonging to the health domain (e.g. AIDS, cancer, infarction) and also to environmental (e.g. climate change), behavioral (e.g. serious car accidents), and technological (e.g. nuclear weapons) domains. The aim has been to investigate the variability in people's judgments about the riskiness of several hazards, with a focus on Covid-19, shedding light on their aversion or indifference to the hazards. Assessing multiple hazards is of great relevance, especially when interested in the risk perception for a new and unknown virus as the Covid-19. In this regard, a study conducted in Italy in

November 2020 [21] explored the risk perception for several diseases, reporting that people had considered more dangerous to catch an unknown disease (the new coronavirus) than well-known diseases such as HIV and influenza.

In our study, an item response theory (IRT) [8,31,32,73] model has been implemented for the analysis of the hazard's attributes that are measured using multiple answer-category items on an ordinal scale, allowing respondents to select a category that they feel to be the most appropriate. Among the considered attributes determining hazard's perceived risk, there are the knowledge of risks from the hazard, media attention, irreversibility of effects, and the harmfulness for children.

We also estimated a latent regression IRT model [22,56,74,77] to examine the role of some individual characteristics (e.g. gender, education, and economic condition), which have been found in the literature [41] to correlate with risk perceptions. Indeed, as also highlighted in Commodari et al. [21], the analysis of the impact of socio-demographics and psychological variables provided greater insight for policymakers on perceived personal and comparative susceptibility to the new coronavirus.

In sum, the paper aimed to improve previous knowledge on the way people perceive the riskiness of Covid-19. Considering multiple hazards and several characteristics allowed us to better understand how the Covid-19 risk perception was framed in the people's cognitive representation of risk perception. The results provided useful advices for developing effective risk communication strategies and control policies oriented to enhance the citizens' awareness and sensitivity and, thus, to favor the use of protective measures.

The work has been organized as follows. In Section 2, we dealt with a detailed description of the survey, the data, and the hazards considered in the analysis, whereas Section 3 has been devoted to a brief review of the statistical models. Section 4 has been focused on the latent trait analysis of the hazards and Section 5 on the relation between Covid-19 risk perception and some individual characteristics. A discussion in Section 6 and some conclusions in Section 7 end the paper.

2. Data

In this section, we illustrated data used in our study. We first described the participants and the procedure adopted to collect data and, then, the characteristics of the self-report questionnaire.

2.1. Participants and procedure

Data collection took place during the first Italian Covid-19 lockdown (March 18, 2020 until May 3, 2020) using an online self-report questionnaire. A large number of respondents has been randomly surveyed and each of them has been invited to recruit future subjects from their acquaintances. The questionnaire was circulated on several social networks (Facebook and Twitter, among others). A sample of $N = 2224$ respondents coming from all Italian regions voluntary took part in the survey and consented to the processing of their data for research purposes. Table 1 shows their socio-demographic characteristics.

Participants were predominantly women (79.4%) with an average age of 31.48 (standard deviation (SD) = 10.83). Most of them reported that they were single (55.00%), employee (42.60%), and with a degree or post-degree qualification (50.30%). About half stated

Table 1. Summary statistics concerning the socio-demographic characteristics of the respondents.

<i>Gender</i>
Female (79.4%); male (20.6%)
<i>Age</i>
Min (18); Max (94); mean (31.48); SD (10.83)
<i>Education</i>
8th grade or lower (4.70%), high school (45.00%), degree/post-degree (50.30%)
<i>Employment</i>
Student (29.30%), working student (12.30%), employee (42.60%), unemployed (14.40%), retired (1.40%)
<i>Marital status</i>
Single (55.00%), married or cohabiting (41.40%), separated/divorced or widowed (3.60%)
<i>Political orientation</i>
Apolitical (30.80%), centrist (4.70%), rightist (10.10%), leftist (27.10%), not responding (27.30%)
<i>Knowledge in the health sector</i>
No (52.20%), Yes, for personal interest (28.80%), Yes, for study/job reasons (19.00%)
<i>Region of residence</i>
North-West (21.80%), North-East (13.90%), Center (18.80%), South (37.50%), Islands (8.00%)

not having knowledge in the health sector (52.20%). Regarding the political orientation, 27.30% did not respond, 30.80% declared to be apolitical, 27.10% leftist, 10.10% rightist, and 4.70% centrist. It is worth noting that the larger proportion of women in the sample is consistent with the greater propensity of females to participate in online surveys than males, as reported by Smith [67].

2.2. Measures of risk perception

Two sections of the self-report questionnaire focused on the assessment of people's risk perception about the considered set of hazards, according to the literature on the psychometric paradigm [28,63]. Within the latter, people are asked to make quantitative judgments about the riskiness of various hazards. These judgments are assumed to reflect people's representation (or 'cognitive map') of risk perception. The set of hazards and characteristics to be considered are selected according to the research purpose. Readers can find many examples of this methodology in the works of Slovic [9,62], who is one of the leading researchers in the field of the psychometric paradigm.

In the first section of the questionnaire, participants were asked to judge both the personal and social riskiness of Covid-19 and other 13 hazards (listed in Table 2) on a 10-point Likert response scale, from 1 = not at all risky to 10 = very risky. As shown in Table 2, the hazards we considered belong to the health, environmental, behavioral, and technological domains. The goal in selecting these hazards was to provide an appropriate context to understand the Covid-19 perceived risk. The selection criteria included prevalence, popularity, and importance of the hazards. For example, diseases of the circulatory system and cancer represent the main causes of death in Italy, as well as the prevalence of diabetes has almost doubled in 30 years (<https://www.istat.it/en/> <https://www.istat.it/en/>). Another example regards environmental hazards such as climate change, which have increasingly worried experts worldwide and also in Italy [70].

Table 2. Risk perception.

Hazard	Median		Mean		SD		$\mathcal{LT}$ index		Cohen's	
	Soc	Per	Soc	Per	Soc	Per	Soc	Per	k	CMH
AIDS	8	7	7.24	6.50	2.54	3.21	0.66	0.69	0.42	945.52*
Cancer	9	9	8.10	8.06	2.32	2.42	0.38	0.36	0.57	1431.50*
Climate change	8	8	7.52	7.33	2.55	2.70	0.56	0.60	0.56	1341.30*
Covid-19	8	7	7.58	6.48	2.38	2.70	0.57	0.84	0.42	1204.70*
Diabetes	7	7	6.68	6.13	2.35	2.83	0.75	0.92	0.40	832.53*
Drug use	8	8	7.32	6.42	2.52	3.35	0.63	0.66	0.39	730.70*
Infarction	8	8	7.48	7.04	2.40	2.77	0.60	0.68	0.42	876.25*
Influenza	4	3	4.14	3.37	2.12	2.16	0.70	0.59	0.45	997.26*
Nuclear weapons	9	8	7.26	6.82	3.01	3.27	0.44	0.52	0.55	1564.80*
Serious car accidents	8	9	7.49	7.66	2.48	2.56	0.59	0.48	0.48	1263.80*
Serious work accidents	8	8	7.56	6.91	2.46	3.06	0.57	0.63	0.45	937.28*
Smoking	8	8	7.17	7.06	2.57	3.01	0.66	0.57	0.46	836.04*
Terrorist acts	8	8	7.12	6.61	2.86	3.23	0.58	0.63	0.53	1519.50*
Vaccines	2	1	2.51	2.38	2.06	2.20	0.30	0.21	0.65	1408.70*

Notes: Median, mean, SD and $\mathcal{LT}$ index for items assessing the global personal (Per) and social (Soc) riskiness on a 10-point Likert scale. The weighted Cohen's kappa and the row-means CMH test computed on personal and social judgment of risk for all the considered hazards ($df = 9$) are reported as a measure of agreement and independence, respectively. In boldface the hazards we focused on.

* p -Value < 0.01.

Table 2 shows the results of people's rating of hazards with regard to the riskiness for themselves (columns Per) and for the Italian society as a whole (columns Soc). According to the median, Covid-19 riskiness is globally comparable to that of AIDS at both personal and social levels. On the other hand, hazards judged to be riskier than Covid-19, at least for one of the two levels (individual and social), were cancer, nuclear weapons, and serious car accidents. The analysis of positioning of Covid-19 with respect to other hazards has been performed focusing on these four hazards (AIDS, cancer, nuclear weapons, and serious car accidents), as presented in detail in Section 4.

The heterogeneity, that is the pairwise diversity between the units, is computed by means of the normalized Laakso and Taagepera [40] index (Table 2, columns $\mathcal{LT}$), $\mathcal{LT} = (1/(m-1))[(\sum_{y=1}^m f_y^2)^{-1} - 1]$, $y = 1, \dots, m$, where f_y are the observed relative frequencies and m denotes the number of ordinal categories for each hazard. The $\mathcal{LT}$ index increases if the distribution of the response categories of the hazard tends to the probability mass function of the discrete Uniform random variable. In other words, high values of the $\mathcal{LT}$ index denote diversity of respondents' perceived risk that, in making a choice of the available response categories, tend to provide responses very different for each other. This situation adds an extra variability to the frequencies (i.e. heterogeneity) generating a distribution with a basic level of uncertainty which spreads around all categories (see Ref. [14] and references therein). Looking at Table 2 (columns $\mathcal{LT}$), we noted that, at society level, opinion of respondents about the riskiness of Covid-19 presented an intermediate level of heterogeneity ($\mathcal{LT} = 0.57$ for Covid-19; minimum: $\mathcal{LT} = 0.30$ for vaccines and $\mathcal{LT} = 0.38$ for cancer; maximum: $\mathcal{LT} = 0.75$ for diabetes and $\mathcal{LT} = 0.70$ for influenza). On the other hand, at personal level, a very wide expression of opinions about riskiness of Covid-19 stood out, as outlined by $\mathcal{LT}$ index that equals 0.84, second only to diabetes ($\mathcal{LT} = 0.92$).

Furthermore, to assess the strength of agreement between personal and social judgment on perceived risk, we computed the weighted Cohen's k [19] for all the considered hazards (Table 2, column Cohen's k). According to the qualitative categorization of Cohen's k values proposed by Landis and Koch [43], we concluded in favor of a moderate agreement for all the hazards (k values between 0.42 and 0.57), included Covid-19 ($k = 0.42$) and except for vaccines, which report a substantial agreement ($k = 0.65$), and diabetes and drug use, for which the agreement is fair ($k = 0.40$ and $k = 0.39$, respectively). To strength these results and to assess whether the rating distributions Per were different from the ratings Soc of each hazard, the row-means Cochran–Mantel Haenszel test (CMH; Table 2, last column) [18,42,48–50] has been also computed. It allows for detecting (latent or manifest) location shifts in the conditional row distributions related to a contingency table. The row-means CMH statistic is designed to test the null hypothesis of no association against the alternative that the row mean scores differ, on average, across strata. Results in Table 2 show statistical evidence of a shift in locations in the row distributions.

After the above-mentioned general questions about riskiness leading to a first ranking of the hazards, the second section of the questionnaire focused on several aspects of the same hazard. Here, respondents compared their perceived trait level with the item content [2]. In particular, participants judged each of the 14 hazards with regard to eight attributions (see Table 3), answering to as many items on a 7-point Likert response scale (from 1 = not at all to 7 = a great deal). Some of these aspects were selected because of their relevance in other studies about risk perception (e.g. knowledge, control), whereas others were selected because of their potential importance in the particular context we considered (e.g. trust in institutions, attention given by media). It is worth noting that all the 14 hazards were rated on the same aspect before the next one was considered.

Finally, a further section aimed to measure the following variables we hypothesized to correlate with Covid-19 risk perception, in addition to the socio-demographic variables described in Table 1:

- Economic condition ('Do you consider your household income sufficient to see the family through to the end of the month?', 10-point Likert response scale from 1 = with great difficulty to 10 = very easily).
- Interest for news about Covid-19 ('How often have you been interested in the Covid-19 news during this period?', 7-point Likert response scale from 1 = not at all to 7 = a great deal).

Table 3. Risk perception.

Item	Description
Item 1	Knowledge of risks from the hazard
Item 2	Risk of long-term effects of the hazard
Item 3	Attention given by media to the risks from the hazard
Item 4	Trust in institutions involved in the management
Item 5	Irreversibility of effects
Item 6	Harmfulness for children
Item 7	Fear caused by the hazard in your own peers
Item 8	Control over the potential damages due to the hazard

Note: Description of hazards' attributions.

- Agreement with the Government guidelines ('Do you agree with the Government guidelines?', 7-point Likert response scale from 1 = not at all to 7 = a great deal).
- Application of Government guidelines ('Do you apply the Government guidelines?', 7-point Likert response scale from 1 = occasionally to 7 = carefully).

3. IRT models for Likert-scale data

In this section, we dealt with the polytomous IRT model developed for item responses characterized by ordered categories. The Samejima's graded response model (GRM; [57]), which can also be included in the generalized linear latent variable models framework [6], represented the reference method we used for the measurement of the hazard's perceived risk. As an alternative to GRM, other models based on the unidimensionality assumption and, usually, on the assumption of the normality for the latent trait have been introduced: partial credit model [51], rating scale model [2], generalized partial credit model [52], and sequential-type model [72] are just some of the examples; among others, see Bartolucci et al. [8] for details on the most commonly used IRT models. In this contribution, we focused on GRM after a selection obtained by comparing likelihood-ratio tests for nested models and information criteria (mainly, the Bayesian Information Criterion, BIC [58]) for non-nested ones.

3.1. The graded response model (GRM)

Let θ_i be an unobservable (*latent*) variable denoting the risk for a certain adverse situation (*hazard*) perceived by individual i ($i = 1, \dots, n$). As usual in the IRT context, we assumed that multiple items contributed to measure latent variable θ_i . Thus, let Y_{ij} denote the response of individual i to item j ($j = 1, \dots, J$) and y_{ij} the observed value of Y_{ij} , with $y = 1, \dots, m_j$. In general, each item j may have a specific number m_j of response alternatives; in our case study, a simplified version has been occurred with a fixed number of categories $m_j = m$ for all the j items, thus $y = 1, \dots, m$.

In the GRM, the probability of individual i with perceived risk θ_i to sign a score y to the item j , $P_{jy}(\theta_i) = P(Y_{ij} = y|\theta_i)$, known also as item response category characteristic curve (IRCCC), is expressed as

$$P_{jy}(\theta_i) = P_{jy}^*(\theta_i) - P_{j,y+1}^*(\theta_i), \quad (1)$$

where $P_{jy}^*(\theta_i) = P(Y_{ij} \geq y|\theta_i)$ represents the probability of an item response Y_{ij} falling in or above the score y , conditional on the latent trait level θ_i . This model has been defined as 'difference' model because the probabilities are given as differences [71].

Formally, $P_{jy}^*(\theta_i)$ is formulated through the global logit link function, that is

$$P_{jy}^*(\theta_i) = \frac{\exp[\alpha_j(\theta_i - \beta_{j,y-1})]}{1 + \exp[\alpha_j(\theta_i - \beta_{j,y-1})]}, \quad y = 2, \dots, m, \quad (2)$$

with α_j item-specific *discrimination* parameter and $\beta_{j,y-1}$ item-threshold specific *difficulty* parameter. Parameter α_j denotes how strongly the item discriminates among different levels of the latent trait, whereas $\beta_{j,y-1}$ points out the *difficulty* of passing from answering category $y-1$ or smaller to answering category y or higher to item j . No constraint of

equal distance between $\beta_{j,y-1}$ parameters is required; they are only constrained to have an increasing order $\beta_{j1} < \dots < \beta_{j,m-1}$. Note that, for each item one discrimination parameter and $m-1$ difficulty parameters are estimated.

The computation of probability in Equation (1) requires two steps. The first one consists in the computation of m probabilities according to Equation (2) [27], while the second one is the computation of the IRCCCs through Equation (1) for all the m response categories within any item j .

The IRCCCs resulting from the two-step process are non-monotonic curves, except for the first and last response categories that follow monotonic decreasing and monotonic increasing logistic trends, respectively. Parameters $\beta_{j,y-1}$ lead the location of the IRCCCs along the latent continuum. In particular, the mode of the non-monotonic IRCCCs is given by the midpoint of two adjacent item-threshold difficulty parameters, whereas the inflection points of the first and the last IRCCCs are detected by β_{j1} and $\beta_{j,m-1}$, respectively. More in detail, the smallest difficulty parameter (β_{j1}) denotes the level of perceived risk for which the probability of answering the smallest category is 50% (i.e. $P(Y_{ij} = 1|\theta_i) = 0.50$) and, similarly, the largest difficulty parameter ($\beta_{j,m-1}$) denotes the level of perceived risk for which the probability of answering the highest category is 50% (i.e. $P(Y_{ij} = m|\theta_i) = 0.50$). To make easier the interpretation of parameters, keep in mind that items with IRCCCs shifted to the right side of the latent continuum imply a *high* average value of difficulties $\beta_{j,m-1}$ and a *high* value of β_{j1} , thus detecting aspects for which individuals tend to select low categories, even when the perceived risk for the hazard is high. On the opposite, items with IRCCCs shifted to the left side of the latent continuum imply a *small* average value of difficulties $\beta_{j,y-1}$ and a *small* value of $\beta_{j,m-1}$, thus detecting aspects for which individuals tend to select high categories, even when the perceived risk for the hazard is low.

To synthesize the amount of discrimination and difficulty provided by an item, the item information curve (IIC) may be considered. A general formulation of the Fisher information of an item can be expressed as

$$I_j(\theta_i) = \sum_{y=1}^m \frac{P'_{jy}(\theta_i)^2}{P_{jy}(\theta_i)}, \quad (3)$$

where $P'_{jy}(\theta)$ is the first derivative of the IRCCC $P_{jy}(\theta_i)$ evaluated at a specific latent trait level. IICs are additive across items that are calibrated on a common latent scale. Samejima [57] showed that the item information increases if ordinal responses are split into more categories. Therefore, the overall Fisher information of the test may be obtained by summing the information of the single items given by Equation (3).

An IIC is usually bell shaped and its shape depends on the item parameters. It reaches its maximum value at the level of the perceived risk corresponding to the item discrimination parameter α_j . Therefore, the item information decreases as the perceived risk departs from the item discrimination and approaches zero at the extremes of the perceived risk scale. Moreover, the difficulty parameters $\beta_{j,y-1}$ influence the location of the curve along the latent continuum, with low values of difficulty providing information for low latent variable values and high values of difficulty yielding information for high values of the latent variable.

3.2. The latent regression GRM

Several extensions of traditional GRM have been proposed in the literature in order to overcome some restrictive assumptions and to make the model more flexible. Among the main contributions, we remind [22,56,74,77] where an explanatory item response modeling approach has been discussed. Aim of these extensions is to identify the determinants of the latent variable and estimate their effects. Indeed, when information is available about the participants taking a survey, it might be interesting to estimate the extent to which the latent variable θ_i is affected by the respondent characteristics. Thus, it is possible to identify a measurement sub-model, such as the GRM in (2), and an explanatory (*structural*) sub-model achieved by constructing a (multiple) regression model for latent variable θ_i . Formally, in the vein of multiple indicators multiple causes framework [46], we set in Equation (2)

$$\theta_i = \mathbf{x}_i' \boldsymbol{\gamma} + \epsilon_i, \quad i = 1, \dots, n, \quad (4)$$

with $\boldsymbol{\gamma}$ vector of regression coefficients describing the effect of covariates $\mathbf{x}_i$ for explaining the latent trait and ϵ_i error component that is usually assumed normally distributed, $\epsilon_i \sim N(\mu, \sigma^2)$. Substituting Equation (4) in Equation (2), $P_{jy}^*(\theta_i)$ is now a function of the observable $\mathbf{x}_i$ and the unobservable ϵ_i , thus

$$P_{jy}^*(\theta_i) = \frac{\exp[\alpha_j(\mathbf{x}_i' \boldsymbol{\gamma} - \beta_{j,y-1} + \epsilon_i)]}{1 + \exp[\alpha_j(\mathbf{x}_i' \boldsymbol{\gamma} - \beta_{j,y-1} + \epsilon_i)]} = P_{jy}^*(\epsilon_i)$$

and

$$P_{jy}(\epsilon_i) = P_{jy}^*(\epsilon_i) - P_{j,y+1}^*(\epsilon_i),$$

where $P_{jy}^*(\epsilon_i) = P(Y_{ij} \geq y | \epsilon_i)$ and $P_{jy}(\epsilon_i) = P(Y_{ij} = y | \epsilon_i)$. In such a context also non-linear, multilevel, or longitudinal structures for the explanatory sub-model and item-by-category covariates for the measurement sub-model may be considered [56].

It is worth to outline that in the latent regression IRT modeling, the estimation of the item parameters α_j and β_{jy} (from the measurement sub-model) and the regression coefficients $\boldsymbol{\gamma}$ (from the explanatory sub-model) happens in a single step. This is different from the two-step approach, consisting in first estimating an IRT model and, then, using the estimates of the latent traits as observed responses in a regression model. The two-step approach is computationally easier to be implemented, but estimators of regression coefficients suffer from inconsistency. Differently, the latent regression approach guarantees the consistency of estimates [22].

3.3. Statistical inference

Estimation of model parameters is based on the maximum likelihood approach. In the IRT literature, there have developed three major methods, namely conditional, full, and marginal maximum likelihood. A detailed overview of these methods is presented in Agresti [1], Baker and Kim [4], Bartolucci et al. [8]. In addition, parameter and ability estimation under a Bayesian approach is reviewed in Baker and Kim [4], among others. Whenever the latent variables are treated as random, as usual in the latent regression IRT framework [56], the marginal maximum likelihood estimation is considered. It is based on maximizing the log-likelihood obtained by integrating out the latent variables.

Let us denote by $\mathbf{y}_i = (y_{i1}, y_{i2}, \dots, y_{ij}, \dots, y_{ij})'$ the response vector of individual i to item j . The definition of the likelihood function relies on the manifest distribution of $\mathbf{y}_i$

$$P(\mathbf{y}_i) = \int_{\mathbb{R}} P(\mathbf{y}_i | \epsilon_i) \phi(\epsilon_i; \mu, \sigma^2) d\epsilon,$$

where $\phi()$ is the density function of the normal distribution, and in virtue of the local independence assumption, $P(\mathbf{y}_i | \epsilon_i) = \prod_{j=1}^J P_{jy}(\epsilon_i)$. Let δ be the vector of free model parameters (i.e. item parameters α_j and β_{jy} , regression coefficients $\boldsymbol{\gamma}$, and parameters μ and σ^2 of the normal distribution of ϵ_i). The likelihood function is then defined as $\mathcal{L}(\delta) = \prod_{i=1}^n P(\mathbf{y}_i)$ and the corresponding log-likelihood function as

$$\ell(\delta) = \sum_{i=1}^n \log P(\mathbf{y}_i) = \sum_{i=1}^n \log \int_{\mathbb{R}} \prod_{j=1}^J P_{jy}(\epsilon_i) \phi(\epsilon_i; \mu, \sigma^2) d\epsilon. \quad (5)$$

The integral contained in $P(\mathbf{y}_i)$ (and then in $\ell(\delta)$) has no solution in closed form. Basically, there exist three approaches to treat this integral: (i) approximating it with numerical integration techniques (e.g. Gauss–Hermite quadrature); (ii) approximating the nonlinear model with a linear one, so that algorithms for linear mixed models may be used to estimate the model parameters; and (iii) adopting Bayesian procedures. In the frequentist setting, solution (i) is usually adopted and the approach of Bock and Lieberman [11] and Bock and Aitkin [10] is implemented.

Following Bock and Aitkin [10] and Bock and Lieberman [11], we first apply the Gauss–Hermite quadrature method to replace the integral in $P(\mathbf{y}_i)$ by a finite sum

$$P(\mathbf{y}_i) \cong \sum_{v=1}^k P(\mathbf{y}_i | \xi_v) \pi_v = \sum_{v=1}^k \left[\prod_{j=1}^J P_{jy}(\xi_v) \right] \pi_v, \quad (6)$$

where $\xi_1, \dots, \xi_k$ are the quadrature points (nodes) and $\pi_1, \dots, \pi_k$ are the corresponding weights. Under the normality assumption of ϵ_i , nodes are usually taken in the interval $[-4; +4]$ [3] and weights are obtained as normalized densities, that is,

$$\pi_v = \frac{\phi(\xi_v)}{\sum_{u=1}^k \phi(\xi_u)}.$$

Second, we maximize the log-likelihood function obtained substituting Equation (6) in Equation (5), that is,

$$\ell(\delta) = \sum_{i=1}^n \log \sum_{v=1}^k P(\mathbf{y}_i | \xi_v) \pi_v = \sum_{i=1}^n \log \sum_{v=1}^k \left[\prod_{j=1}^J P_{jy}(\xi_v) \right] \pi_v.$$

Function $\ell(\delta)$ may be maximized by direct (e.g. Newton–Raphson algorithm) or indirect (e.g. expectation-maximization (EM) algorithm; [23]) methods. Alternatively, it can be adopted a quasi-likelihood approach, a Bayesian approach based on Markov chain Monte Carlo methods, or a semi-parametric approach. Following Bock and Aitkin [10], the most common approach is based on the EM algorithm whose details are confined in the Appendix.

The standard errors of model parameters are obtained as the square root of the reciprocal value of the information matrix. Moreover, once the model parameters have been estimated, the values of the latent variables for each respondent can be estimated by treating item parameters as known and maximizing the log-likelihood with respect to the latent trait or, alternatively, using the expected value or the maximum value of the corresponding posterior distribution in a Bayesian perspective. See Bacci et al. [3] and Bartolucci et al. [7] for a detailed description.

4. Measurement of perceived risk: Covid-19 vs. other hazards

The perceived risks for the hazards at issue (i.e. Covid-19, AIDS, cancer, nuclear weapons, and serious car accidents) were measured relying on the observed responses to items 1–8 and comparing several alternative IRT models. Based on the BIC index, the best model resulted the GRM equation (1), independently of the type of hazard (results are available from authors upon request). All the models have been implemented using the *mirt* package [16] of R version 4.0.2 [55].

Overall, the test provided a satisfactorily Fisher information for any hazard. Indeed, the test information curves embraced a wide range of the latent traits, as shown in Figure 1 (left panel) for the Covid-19; plots of the other hazards are similar (Figure 2).

Looking at information of single items (Figure 1, right panel), item 2 (long-term effects) has been the most informative for the Covid-19, followed by item 5 (irreversibility of effects) and item 6 (harmfulness for children). The other hazards presented some differences (Figure 3), mainly as concerns item 6, which has been the most informative item (together with item 2) for nuclear weapons and serious car accidents, and item 1 (knowledge of risks), which presented higher levels of information for all the other hazards in comparison with Covid-19.

On the opposite, the information curve for item 8 (control over the potential damages) resulted definitely flat, both for Covid-19 and for the other hazards. In general, individuals assessed to have no control at all over the potential damages due to the hazards, thus item 8 did not provide any contribution to measure the perceived risks and, for this reason, it has been ignored below.

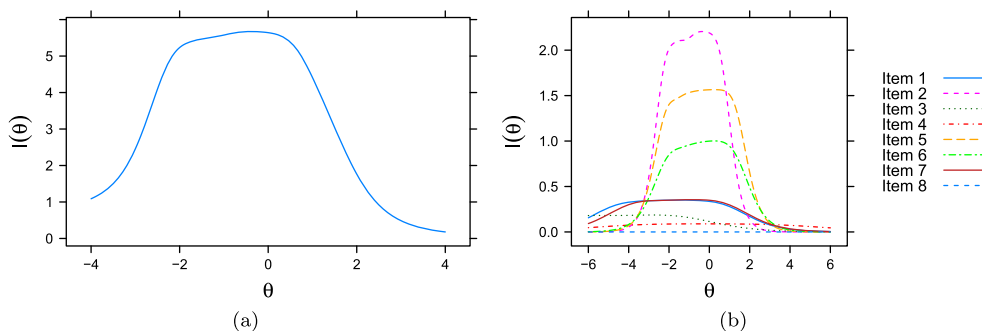


Figure 1. Test and IICs for Covid-19. (a) Test information (Covid-19). (b) Item information (Covid-19).

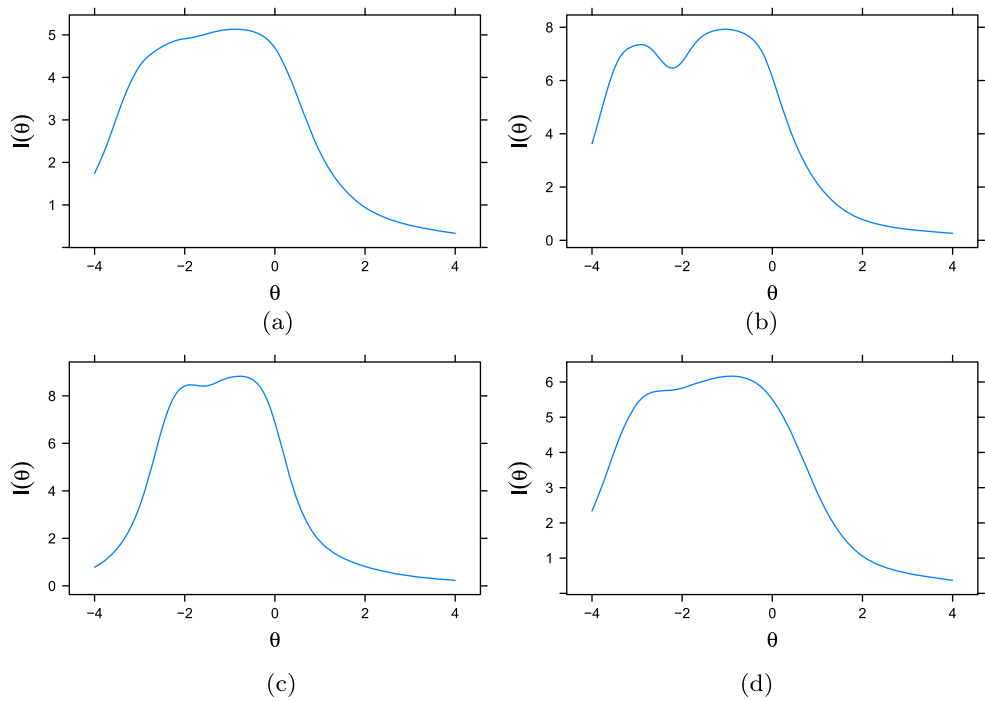


Figure 2. Test information curves for AIDS, cancer, nuclear weapons, and serious car accidents. (a) AIDS. (b) Cancer. (c) Nuclear weapons. (d) Serious car accidents.

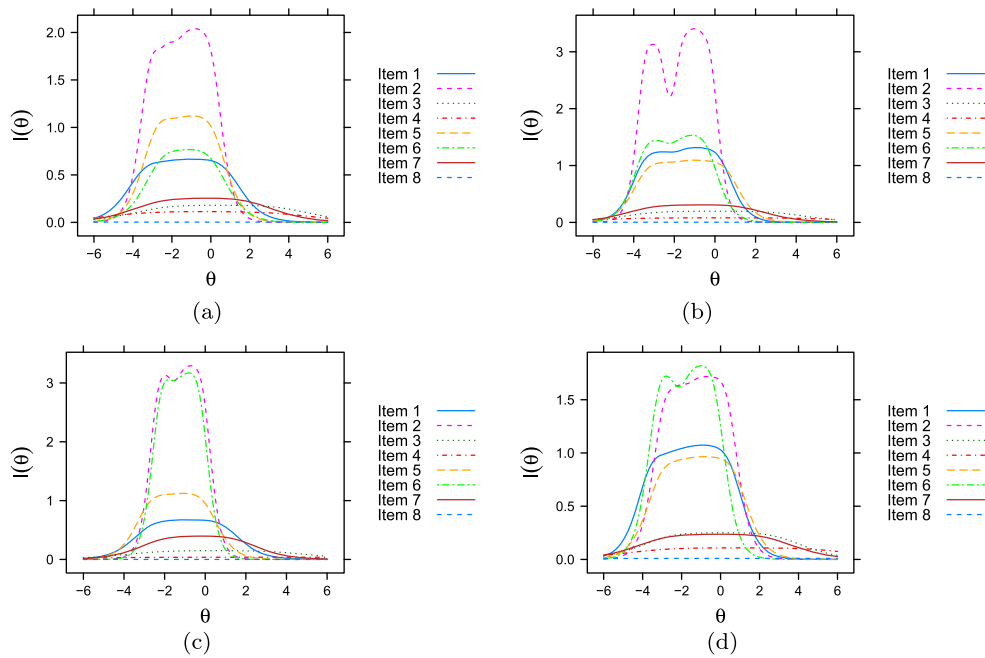


Figure 3. IICs for AIDS, cancer, nuclear weapons, and serious car accidents. (a) AIDS. (b) Cancer. (c) Nuclear weapons. (d) Serious car accidents.

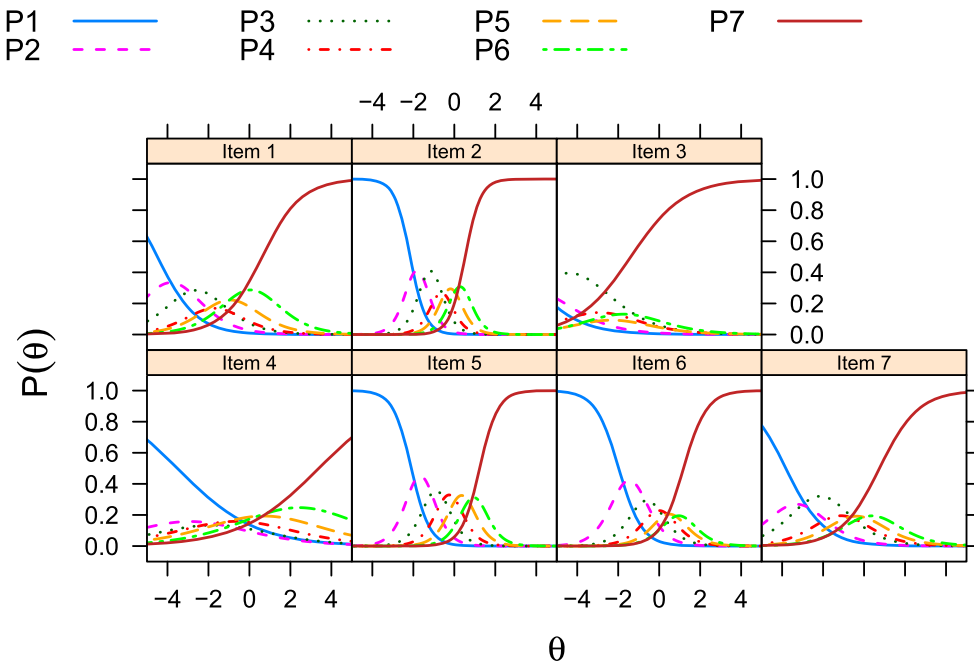


Figure 4. Item-threshold difficulty parameters, by item: $P1 = P(Y_{ij} = 1|\theta_i), \dots, P7 = P(Y_{ij} = 7|\theta_i)$.

Table 4. Average values of $\hat{\beta}_{jy}$ parameters ($j = 1, \dots, 7$), by hazard (in logit).

Item	Covid-19	AIDS	Cancer	Nuclear weapons	Serious car accidents
1. Knowledge of risks	-1.825	-1.154	-1.425	-0.972	-1.439
2. Long-term effects	-0.720	-1.281	-1.626	-1.263	-1.183
3. Attention by media	-3.636	0.614	0.051	0.400	0.072
4. Trust in institutions	-0.382	-0.277	0.073	2.364	0.890
5. Irreversibility of effects	-0.321	-1.260	-1.302	-1.463	-1.124
6. Harmfulness for children	-0.164	d1.115	-1.685	-1.294	-1.641
7. Fear in peers	-1.346	-0.142	-0.769	-0.357	-0.150

To better understand what specific aspects provided the most relevant contribution to the perceived risk for Covid-19, it has been useful to analyze the IRCCCs, whose plots are shown in Figure 4.

As outlined in Section 3.1, the location of the IRCCCs along the latent continuum has been driven by the estimated item-threshold difficulty parameters $\hat{\beta}_{j,y-1}$ ($y = 2, \dots, 7$) that are always ordered ($\hat{\beta}_{j1} < \hat{\beta}_{j2} < \dots < \hat{\beta}_{j6}$). To make easier the evaluation of the IRCCCs, we focused on the averages of $\hat{\beta}_{j,y-1}$ ($y = 2, \dots, 7$) and on values of $\hat{\beta}_{j1}$ and $\hat{\beta}_{j6}$, comparing Covid-19 vs. the other hazards. The average difficulties of items 1–7 are displayed in Table 4, whereas points $(\hat{\beta}_{j6}, \hat{\beta}_{j1})$ are plotted in Figure 5. To interpret output displayed in Figure 5, we recall that a high average value of parameters $\hat{\beta}_{j1}, \dots, \hat{\beta}_{j6}$ as well as a high value of $\hat{\beta}_{j1}$ denote aspects for which individuals tend to select low response categories, even when the perceived risk for the hazard is high. Furthermore, a small average value of $\hat{\beta}_{j1}, \dots, \hat{\beta}_{j6}$ as well as a small value of $\hat{\beta}_{j6}$ detect aspects for which individuals tend to select high response categories, even when the perceived risk for the hazard is low.

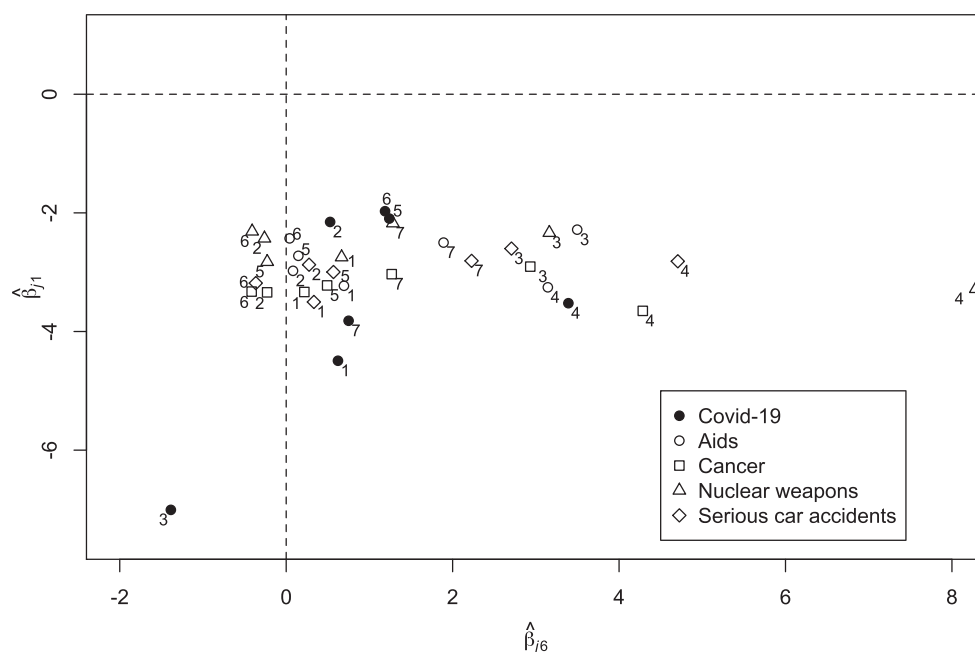


Figure 5. Item-threshold difficulty parameters, by hazard: $\hat{\beta}_{j1}$ (y-axis) vs. $\hat{\beta}_{j6}$ (x-axis) ($j = 1, \dots, 7$).

Looking at Table 4 (column Covid-19), the most critical aspect has been the attention given by media to the risks related with Covid-19 (item 3; average of $\hat{\beta}_{jy}$ equal to -3.636); then, the knowledge of risks (item 1) and the fear caused by Covid-19 in the peers (item 7) followed.

Comparing Covid-19 with the other hazards, aspects perceived as definitely less dangerous were: the long-term effects (item 2), the irreversibility of the effects (item 5), and the harmfulness for children (item 6): namely, both $\hat{\beta}_{j1}$ and $\hat{\beta}_{j6}$ parameters (see Figure 5) as well as the averages of $\hat{\beta}_{jy}$ of Covid-19 were higher (lower in absolute value) than the other hazards. On the opposite, the perceived fear caused by Covid-19 (item 7) is higher with respect to the other hazards. As concerns the trust in institutions (item 4), all the hazards (in particular, nuclear weapons) presented low levels of trust (high values of $\hat{\beta}_{j6}$): Covid-19 stayed in an intermediate position, with individuals that tend to have more trust in institutions for the management of Covid-19 than for the management of nuclear weapons, cancer, and serious car accidents; the level of trust was similar to that in the management of AIDS. Finally, the positioning with respect to item 1 (knowledge of risks) is similar to the other hazards for high levels of perceived risks, but overall (compare averages of $\hat{\beta}_{jy}$ in Table 4) individuals stated a superior knowledge of associated risks.

After estimating item parameters of the GRM equation (1), the values of the latent variable θ_i have been estimated for each individual i , as outlined in Section 3.3. Figure 6 displays the distributions of estimated perceived risks for Covid-19 and the other hazards, together with the correlation coefficients between pair of hazards. Perceived risk for Covid-19 resulted significantly positively correlated with the other perceived risks at an intermediate extent (correlation coefficients around 0.50), being the highest correlations

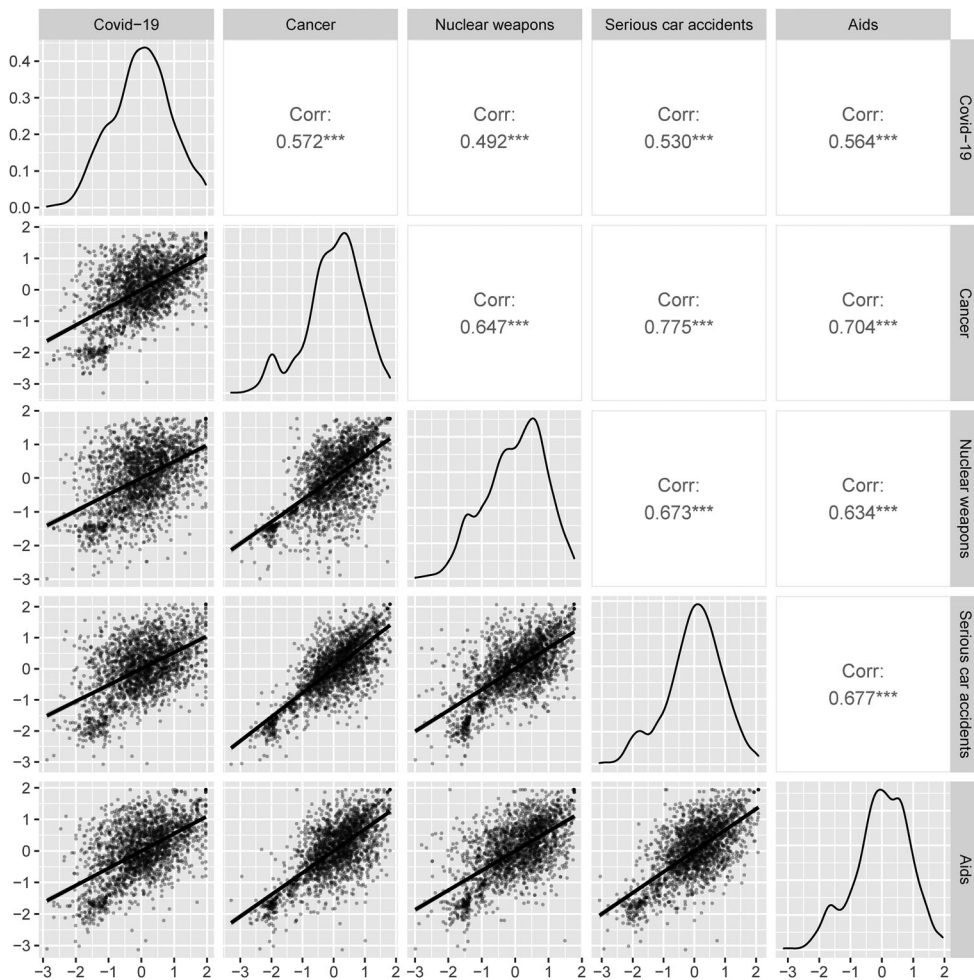


Figure 6. Distribution of perceived risks and correlations between perceived risks, by hazard.

observed between perceived risks for cancer and serious car accidents (0.775) and cancer and AIDS (0.704).

5. Latent regression analysis of perceived risk for Covid-19

To investigate the effect of covariates on the perceived risk for Covid-19, we estimated a latent regression GRM (Section 3.2). As expected, the estimates of item parameters are the same as those discussed in the previous section, because the measurement part of the model is unchanged. In addition, the explanatory sub-model equation (4) provided insights on the effects of individual characteristics. Table 5 shows the estimates of regression coefficients, together with standard errors, z -statistics, p -values, and inferior and superior limits of confidence intervals at 95%. Displayed results refer to the selected model containing only covariates whose estimated coefficients γ were significant (significance level at 5%).

The perception of risk for Covid-19 resulted to be significantly lower for males (vs. females), married or cohabitant people (vs. single, separated/divorced people, or

Table 5. Latent regression analysis: regression coefficients (coeff.), standard errors (s.e.), z-statistics, *p*-values, and inferior (Inf.) and superior (Sup.) limits of 95% confidence intervals (only statistically significant covariates at 5% level).

Covariate	Coeff.	s.e.	z-Stat	<i>p</i> -Value	Inf.	Sup.
Male	−0.307	0.056	−5.443	0.000	−0.418	−0.197
Married	−0.142	0.055	−2.584	0.010	−0.249	−0.034
Student	−0.194	0.056	−3.473	0.001	−0.303	−0.085
Retired	0.404	0.199	2.034	0.042	0.015	0.793
Leftist	−0.223	0.054	−4.152	0.000	−0.328	−0.118
Economic condition	−0.060	0.011	−5.494	0.000	−0.082	−0.039
Living in Center of Italy	−0.137	0.061	−2.233	0.026	−0.258	−0.017
Living in North-East of Italy	−0.224	0.069	−3.263	0.001	−0.359	−0.090
Interest for news about Covid-19	0.119	0.020	5.982	0.000	0.080	0.158
Application of Government guidelines	−0.099	0.023	−4.24	0.000	−0.144	−0.053
Agreement with Government guidelines	0.138	0.017	8.306	0.000	0.105	0.170

widowers), students (vs. workers), leftists (vs. people with other political orientation), and people living in the Center and in the North-East of Italy (vs. people living in the South, Islands, and North-West), whereas retired people presented higher level of perceived risk; moreover, wealthier people tended to have a lower perceived risk. In addition, we also controlled for the interest for news from media, the agreement with the Government guidelines, and the application of the Government guidelines: high levels of interest for news from media and of agreement with Government guidelines and poor application of the Government guidelines were associated with an increasing perceived risk for Covid-19. Finally, it is worth to outline that no significant association resulted for age, education level, and specific knowledge in the health sector.

6. Discussion

The contribution provided a useful application of latent trait models in the field of risk assessment, allowing to identify the main factors (hazard's characteristics and individual covariates) affecting public perception of risk. Risk perception plays a central role in determining individuals' responses to the hazards and is a considerable precursor to protective behavior, as several models of behavior change have highlighted [53]. The Covid-19 pandemic represented a unique point of view in this research area due to its worldwide emergency and its substantial effects on people's lives.

According to the psychometric paradigm, in our study, people first made quantitative judgments about the perceived personal and social riskiness of Covid-19 and other 13 hazards. Results showed that Covid-19 riskiness is globally comparable to that of AIDS at both personal and social levels. Cancer, nuclear weapons, and serious car accidents were instead judged to be riskier than Covid-19, at least for one of the two levels (individual and social). On the other hand, people deemed vaccines as the least dangerous, followed by influenza and diabetes.

Looking at the heterogeneity index, the respondents overall had lower indecision in making a judgment about the riskiness of cancer, nuclear weapons, serious car accidents, and vaccines. Moreover, more uncertainty was found in personal riskiness assessment than the social one for Covid-19, diabetes, infarction, and nuclear weapons. On the opposite, greater indecision in social riskiness judgment emerged for influenza, serious car accidents, smoking, and vaccines.

The comparison between personal and social judgment has greatly interested researchers in the context of risk perception, shedding light on the well-known optimistic bias or unrealistic optimism [76], namely the tendency to believe that hazards are less risky for oneself than for other people ('It won't happen to me'). We measured the strength of agreement and association between personal and social judgment through the weighted Choen's k and the row-means CMH test. Results brought out a moderate and significant agreement for all the hazards, pointing out a weaker effect of the optimistic bias on the participants' judgments.

A more in-depth analysis of hazard's attributes exploiting latent trait models allowed to exploring what specific aspects provided the most relevant contribution to the perceived risk for Covid-19 and if there were some relevant characteristics in common with the other hazards.

Overall, the test supplied a satisfactorily Fisher information for any hazard, embracing a wide range of latent traits (i.e. perceived risks from the hazard). More specifically, hazard's attributes that mostly contributed to discriminate the level of perceived risk from the hazards were the long-term effects, the irreversibility of effects, and the harmfulness for children. Thus, differences in perceived risks level imply differences in how people have represented the future scenarios due to the hazard [41], how its effects have been perceived as irreversible, and whether the children have been considered involved [65]. These characteristics proved to be relevant for all the five risks we considered.

In particular, item parameters showed that Covid-19 effects were perceived as less long-terming, irreversible, and harmful for children than the effects of other hazards. About the interpretation of this result, we can speculate that people judgment were influenced by the almost no impact of Covid-19 on children health. Furthermore, the lockdown protective measures in the first-wave of the pandemic appeared to be resolute, affecting people belief about short-term and reversibility of the Covid-19 effects.

Some differences also emerged for the information provided by the knowledge of risks from the hazard, which is lower for all the other hazards in comparison with Covid-19. Looking at item parameters and results about the attention given by the media helped us to understand this difference. Indeed, people stated their superior knowledge of Covid-19 risks as well as greater media attention. This concurrence invokes the heuristic process known as availability bias [37]. People tend to judge an event as likely or frequent if it is easy to imagine or recall. In this vein, widespread media attention increased the availability of information about Covid-19, also causing an increase in people's perceived knowledge about its riskiness. However, media often provided confusing and contradictory information, making the development of true knowledge difficult. We make guess that for this reason, knowledge of risks has had a poor ability to discriminate different levels of Covid-19 risk perception, as also reported in Lanciano et al. [41].

Moreover, the perceived fear caused by Covid-19 in the peers is higher with respect to the other hazards, we hypothesize as an effect of the social amplification of Covid-19 riskiness [38] due to the greater attention given by the media [41].

As regards the trust in institutions involved in risk management, all the hazards presented low levels of trust. In particular, Covid-19 resulted in an intermediate position since individuals stated to have more trust in institutions involved in the management of Covid-19 than in the management of nuclear weapons, cancer, and serious car accidents; the level of trust in institutions involved in the management of AIDS has been

similar to that in the management of Covid-19. In general, risk perception has been positively related to distrust in risk management [60]. Interestingly, our results showed that the level of Covid-19 perceived riskiness raised as people's trust in institutions involved in its management increased.

This unusual association also emerged in another study focusing on Covid-19 risk perception in Italy [41]. In that case, authors invoked the fear appraisal processes to explain the result: trust in the effectiveness of protective measures adopted by the Government led people to support the existence of an objective hazard from which they should protect themselves; this objective risk also sustained perceived subjective risk.

Finally, another critical issue regarded the perceived control over the potential damages due to the hazards. Regardless of perceived risk levels, participants stated to have no control at all both for Covid-19 and for the other hazards, underlying that this characteristic did not provide any contribution to differentiate perceived risks levels.

Besides, the latent regression analysis, focused on the perceived risk for Covid-19, allowed to investigate the role of some individual characteristics. First, Covid-19 perceived riskiness resulted in being significantly lower for males, according to what is known as 'White-Male Effect' in the literature on risk perception [36]. Moreover, married or cohabitant people, students, leftist, and wealthier people tended to have a lower perceived risk whereas retired people – who are generally the most affected – presented high levels of perceived risk. Several studies deepened some of these emerging issues, including the association between sociodemographic characteristics and Covid-19 risk perception [17], the Covid-19 risk perception among college students [15,24], and the role of political ideology in predicting the COVID-19 threat perception and the adherence to the government containment measures [5,13,54].

An interesting result regards differences in people risk perception according to the region of residence. Surprisingly, results showed a lower level of perceived risk in the Center and the North-East of Italy. As the best of our knowledge, we did not find in literature any other results about geographic Italian macro-areas differences in perceived riskiness of Covid-19. Future studies are needed to explore these differences deeply.

On the other hand, no significant association resulted for age, education level, and specific knowledge in the health sector, as also reported in other studies focused on Covid-19 risk perception in Italy [26,41].

In addition, high levels of interest for news from media and of agreement with Government guidelines and poor application of the Government guidelines were associated with an increased perceived risk for Covid-19, in line with the results mentioned above on hazard's characteristics.

7. Conclusions

Several interesting issues emerged in this study, mainly related to practical implications in the context of management and communication of risk.

First, a careful knowledge promotion about the Covid-19 effects and the suitable preventive measures is needed. Risk communications often provided uncertainty and contradictory information, not appropriately improving public knowledge and risk perception. Moreover, the overwhelming amount of worrying information contributed to people's overreaction and negative feelings [33]. This argument invoked the differences

between affective and cognitive risk perception. The former refers to risk as feelings (e.g. worry, dread) and involved heuristic information processing. The latter consists of a cognitive evaluation of risk characteristics and consequences based on analytical information processing [25].

A study on the association between risk perception and well-being during the Covid-19 crisis highlighted that, while affective risk perception negatively impacts public depression and policy support, a better cognitive risk perception increases people's support for prevention policies and reduces people's depression [25]. Hence, to be effective, risk communication should be addressed primarily to improve the cognitive risk perception. Our findings could be of great interest in this vein, providing insights about the hazard and individual characteristics that mainly contribute to producing people's cognitive representation of risk perception. For example, a relevant matter regards the belief of individual controllability on Covid-19 damages. In our study, people stated to have no control over the sanitary emergency, emphasizing the necessity to improve the level of awareness on what people can do to control the Covid-19 spread. In this regard, as reported in Lohiniva et al. [45], a recommendation to enhance risk communication effectiveness consists in shifting the attention from fear messages centered on emergency to contents appealing individual responsibility in adopting the best practices. In fact, the outbreak control mainly depends on people's compliance to the recommended preventive measures, such as the use of face masks and physical distance. A relevant motivation in this sense also comes from peoples' trust in the institutions involved in emergency management. Thus, risk communication should also include constant updates about government actions, effectively providing explanations understandable even by laypeople.

This study has some important strengths. The first resides in the psychometric paradigm principles. Unlike other studies on Covid-19 risk perception, we compared multiple hazards belonging to different domains and considered several hazards' characteristics. Thus, our study is effective in depicting people's cognitive representation of risk perception and, as stated before, in providing some insights to develop compelling risk communication strategies.

On the other hand, the nature of the sample may be one of the drawback of the study. It might be that the most distressed or worried subjects are more likely to participate, known as the concept of symptomatic volunteers. This bias is typical to be a concern in online surveys [68]. Furthermore, we did not assess an 'objective' knowledge that respondents have about Covid-19, an element that predict reduced risk perception [59]. Notwithstanding its limitations, the present study provided insights into how experience, risk perceptions, cultural background, and social norms have affected individual and social risk perception during the first period of the Covid-19 outbreak in Italy.

Future studies will aim at understanding if a sensible reduction of perceived risks will appear when the immunization programs will cover the largest part of Italian population by replicating the survey.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix

A.1. EM algorithm

The EM algorithm relies on the definition of the complete data log-likelihood, which corresponds to the log-likelihood that we could compute if we knew the value of the latent variable for individuals in the sample:

$$\begin{aligned}\ell^*(\delta) &= \sum_{i=1}^n \sum_{v=1}^k \mathcal{I}(\epsilon_i = \xi_v) \log [P(\mathbf{y}_i | \xi_v) \pi_v] \\ &= \sum_{i=1}^n \sum_{v=1}^k \mathcal{I}(\epsilon_i = \xi_v) \log \pi_v + \sum_{i=1}^n \sum_{v=1}^k \mathcal{I}(\epsilon_i = \xi_v) \log \left[\prod_{j=1}^J P_{jv}(\xi_v) \right],\end{aligned}$$

where $\mathcal{I}(\cdot)$ is an indicator function equal to 1 if its argument is true.

The EM algorithm maximizes $\ell^*(\delta)$ alternating two steps until convergence:

E-step. The expected value of $\ell^*(\delta)$ is computed, given the observed data $\mathbf{y}_i$ and the current value of parameters $\hat{\xi}_v$ and $\hat{\pi}_v$. In practice, this expected value corresponds to the posterior probability of ξ_v given the vector of responses $\mathbf{y}_i$, that is,

$$\hat{\mathcal{I}}(\epsilon_i = \xi_v) = P(\xi_v | \mathbf{y}_i) = \frac{P(\mathbf{y}_i | \hat{\xi}_v) \hat{\pi}_v}{\sum_{u=1}^k P(\mathbf{y}_i | \hat{\xi}_u) \hat{\pi}_u}.$$

At the end of E-step, $\hat{\mathcal{I}}(\epsilon_i = \xi_v)$ is substituted in $\ell^*(\delta)$.

M-step. The complete data log-likelihood $\ell^*(\delta)$ is maximized with respect to each element of δ . This step is usually implemented through a Newton–Raphson or a Fisher–Scoring algorithm. Details about the computation of derivatives involved in the application of these algorithms are provided in Baker and Kim [4]; further details are in Bartolucci et al. [7] and Bartolucci et al. [8, Chap. 5], among others.

The E- and M-steps are iterated until the difference in the log-likelihoods of two consecutive steps is lower than a threshold (say 10^{-6}). Note that, as usual with iterative algorithms, we need to initialize

the EM algorithm. To avoid problems of multimodality of the model likelihood, a common strategy consists of using multiple sets of starting values of model parameters that are randomly generated [23]: the class weights are drawn from a continuous uniform distribution between 0 and 1 and then they are normalized so as to sum up to 1; the other parameters are generated from independent standard normal distributions.